

2) Proven

a) $A = A^\dagger$
 $B = B^\dagger$
 $[A, B] = 0$

Show that both A and B have same eigenstate.

$$A|a\rangle = a|a\rangle$$

~~similarly~~

$$\langle a| [A, B] |a\rangle = 0$$

$$\langle a| (AB - BA) |a\rangle = 0$$

$$\langle a| AB |a\rangle - \langle a| BA |a\rangle = 0$$

$$a \langle a| B |a\rangle - \langle a| B |a\rangle a = 0$$

$$(a_i - a_j) \underbrace{\langle a_i| B |a_j\rangle}_{= B_{ij}} = 0$$

$$(a_i - a_j) B_{ij} = 0$$

For $a_i \neq a_j \Rightarrow B_{ij} = 0$

So B has no matrix elements between eigenvectors of A with different eigenvalues

b) Solution

$$[L^2, L_z] = 0 \text{ show this}$$

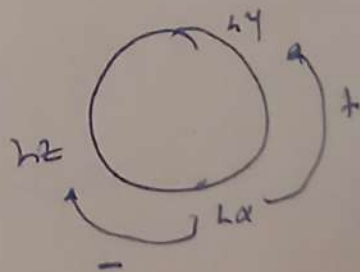
$$[L_x, L_z] = -i\hbar L_y \quad (\text{in } \hbar)$$

$$[L^2, L_z] = [L_x^2 + L_y^2 + L_z^2, L_z]$$

$$= [L_x^2, L_z] + [L_y^2, L_z] + [L_z^2, L_z]$$

$$= L_x [L_x, L_z] + [L_x, L_z] L_x + L_y [L_y, L_z] + [L_y, L_z] L_y + L_z [L_z, L_z] + [L_z, L_z] L_z$$

$$= L_x (-i\hbar L_y) + (-i\hbar L_y) L_x + L_y (i\hbar L_x) + (i\hbar L_x) L_y + 0 + 0$$



$$0Lz + Lz \rightarrow$$

$$= -2iLxLy + 2iLyLx + 0$$

$$= 0 + 0$$

$$= 0$$

$$[L^2, Lz] = 0$$

$\Rightarrow$

c) Solution

$$8) [Lz, L_{\pm}] = [Lz, Lx \pm iLy] = [Lz, Lx] + i[Lz, Ly]$$

$$= [Lz, Lx] \pm i[Lz, Ly] = Ly \pm i(-iLx)$$

$$= \underline{\underline{Ly \pm Lx}}$$

$$= Lx \pm iLy$$

$$= \underline{\underline{\pm L_{\pm}}}$$

Eigenstate $|l, m\rangle$ of L^2 and Lz

$$Lz(L_{\pm}|l, m\rangle) = Lz L_{\pm}|l, m\rangle$$

$$= (L_{\pm} Lz + [Lz, L_{\pm}])|l, m\rangle$$

$$= L_{\pm}(m|l, m\rangle) \pm L_{\pm}|l, m\rangle$$

$$= (m \pm 1)(L_{\pm}|l, m\rangle)$$

//

$$9) [L^2, L_{\pm}] = 0$$

$$[L^2 + Lx^2 + Ly^2, Lx + iLy] = 0$$

$$L^2 = \frac{(L_1^2 + L_2^2 + L_3^2)}{2}$$

$$[L^2, L_{\pm}] = \sum_i (\hbar^2 L_i^0 L_{\pm}^0 + \hbar^2 L_{\pm}^0 L_i^0)$$

$$L^2 (L_{\pm} |l, m\rangle) = L_{\pm} L^2 |l, m\rangle \\ = l(l+1) \hbar^2 (L_{\pm} |l, m\rangle)$$

$$L^2 L_{\pm} |l, m\rangle = l(l+1) \hbar^2 L_{\pm} |l, m\rangle$$

$$L^2 L_{\pm} |l, m\rangle = (m \pm 1) \hbar^2 L_{\pm} |l, m\rangle$$

4) Representation of L^2 in spherical coordinates

$$L = r \times p$$

$$p = -i\hbar \nabla (r = r)$$

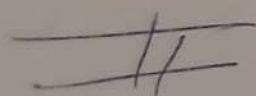
$$L^2 = L^2 - L^2 \\ = -\hbar^2 \left(r^2 \frac{\partial}{\partial r} - r \frac{\partial}{\partial r} \right)$$

Coordinate for spherical shape (r, θ, ϕ)

$$x = r \sin \theta \cos \phi$$

$$y = r \sin \theta \sin \phi$$

$$L^2 = -\hbar^2 \frac{\partial}{\partial \phi}$$



2) Solution

$$F_{nlm}^{(m)} = \langle nlm | W | nlm \rangle = \langle nlm | e^2 F | nlm \rangle$$

$$Z \in F \langle n | m | Z | n | m \rangle$$

Under Parity

$$r \rightarrow 0 - r$$

$$\text{Eigenstate} = |n, m\rangle$$

$$\text{parity} = (-1)^l, \text{ while } Z \text{ is odd}$$

$$\psi_{n, m}^*(r) Z \psi_{n, m}(r) \text{ is odd function}$$

$\rightarrow$ Integral over all space vanishes

$$E_{n, m}^{(1)} = 0 \text{ for particular ground state } |100\rangle$$

$$\int \psi_{1, m}^*(\theta, \phi) \cos\theta \psi_{1, m}(\theta, \phi) d\Omega = 0$$

b) Selection rule: m must be conserved

$$\langle n, m | W | n', m' \rangle = eF \int R_{n, m}(r) R_{n', m'}(r) r^3 dr \int \psi_{1, m}^*(\theta, \phi) \cos\theta \psi_{1, m'}(\theta, \phi) d\Omega$$

$\cos\theta$ has azimuthal quantum number $m=0$

$$\int e^{-im\phi} e^{im'\phi} d\phi \propto \delta_{m, m'}$$

$$\langle n, m | W | n', m' \rangle = 0 \text{ if } m \neq m'$$

c) $n=2$ subspace - diagonalize W

$n=2$, the four state $|200\rangle$ and $|21m\rangle$

$$m = 0, \pm 1$$

Selection Rule

$$\left. \begin{array}{l} \Delta m = 0 \\ \Delta L = \pm 1 \end{array} \right\} \text{Non-zero matrix elements}$$

$|200\rangle$ couples to $|220\rangle$

$|222\rangle$ and $|22, -2\rangle$ are not coupled to $|200\rangle$

$$\langle 200 | W | 220 \rangle = -A \quad A = 3eF_{Q0}$$

$$\langle 220 | W | 200 \rangle = -A$$

$\{ |200\rangle, |220\rangle \}$

$$W_{\text{sub}} = \begin{pmatrix} 0 & -A \\ -A & 0 \end{pmatrix}$$

$\epsilon = \pm A$ with normalized eigenvectors

$$|4\rangle = \frac{1}{\sqrt{2}} (|200\rangle \pm |220\rangle)$$

$$n = 2$$

$$\begin{aligned} E^{(M)} &= \pm A \\ &= \pm 3eF_{Q0} \end{aligned}$$

$$|4\rangle = \frac{1}{\sqrt{2}} (|200\rangle \pm |220\rangle)$$

Two states

$$|2, 2, \pm 2\rangle$$

Undisturbed level

$$E = E_2^{(0)} \pm 3eF_{20}$$

$$E_2^{(0)} = \frac{-e^2}{2a_0(2^2)}$$

$$= \frac{-e^2}{2a_0 \cdot 4}$$

$$= \frac{-e^2}{8a_0}$$